

ALVA BEACH, Australia

WMO: 952960

Lat: 19.4569S

Lon: 147.4833E

Elev: 26

StdP: 14.68

Time Zone: 10.00 (E10)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	51.4	54.3	35.8	30.8	64.2	40.9	37.7	61.7	22.2	70.2	20.7	70.6	7.2	210	0.481

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.4	89.5	78.8	88.2	78.0	87.2	77.4	81.9	85.8	80.8	85.1	79.8	84.4	12.2	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.9	160.6	84.1	79.6	153.9	83.4	78.5	147.9	83.0	45.8	85.7	44.5	85.1	43.5	84.5	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.6	23.0	20.8	46.4	93.4	2.0	3.4	45.0	95.8	43.9	97.8	42.8	99.8	41.3	102.2
			41.5	82.2	1.8	1.7	40.2	83.4	39.2	84.4	38.1	85.4	36.8	86.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	75.4	81.9	82.0	80.2	76.6	71.9	67.8	66.4	67.9	72.9	76.7	79.4	81.7	(6)
	DBStd	6.42	2.32	2.45	2.52	2.66	3.57	4.06	3.98	3.80	3.34	2.91	2.36	2.36	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	70	0	0	0	0	3	20	31	15	1	0	0	0	(9)
	CDD50	9278	989	895	936	797	680	534	510	556	688	829	882	982	(10)
	CDD65	3872	524	475	471	347	218	103	76	106	239	364	432	517	(11)
	CDH74	33714	5971	5456	4967	2808	1138	261	112	256	971	2413	3776	5585	(12)
	CDH80	8375	1965	1863	1395	421	64	2	1	6	45	278	699	1636	(13)
(14) Wind	WSAvg	11.7	12.5	11.9	12.6	11.9	10.1	9.4	9.7	10.4	11.9	13.0	13.8	13.3	(14)
(15) Precipitation	PrecAvg	38.1	8.4	8.9	6.7	2.3	1.7	0.9	0.5	0.7	0.3	1.3	2.1	5.5	(15)
	PrecMax	74.9	41.3	27.9	21.6	15.7	8.6	7.8	2.4	3.9	2.5	9.8	9.0	23.6	(16)
	PrecMin	14.5	0.7	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	(17)
	PrecStd	15.6	8.0	6.4	5.8	3.3	2.0	1.5	0.6	1.0	0.6	2.1	2.4	5.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.7	92.4	90.3	86.8	83.7	80.1	78.1	80.5	83.2	85.5	88.1	89.9	(19)
		MCWB	79.9	79.4	78.3	75.9	73.3	70.8	67.7	67.2	71.1	74.2	76.4	79.2	(20)
	2%	DB	88.9	89.6	88.1	85.0	81.7	78.3	76.5	78.1	81.2	83.8	86.1	88.1	(21)
		MCWB	79.1	78.8	77.7	75.1	71.8	70.0	67.3	67.8	70.8	74.3	75.8	77.8	(22)
	5%	DB	87.6	88.1	86.8	83.7	80.3	76.9	75.3	76.6	80.0	82.7	84.8	87.1	(23)
		MCWB	78.1	78.1	77.1	74.3	70.9	69.0	66.6	67.4	70.2	73.2	74.8	77.0	(24)
	10%	DB	86.5	87.0	85.5	82.5	79.0	75.5	74.1	75.3	78.7	81.6	83.6	86.0	(25)
		MCWB	77.2	77.7	76.4	73.6	70.5	67.9	66.0	66.7	69.6	72.4	73.9	76.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.7	83.7	81.7	79.6	76.5	74.8	71.8	72.7	75.3	78.9	81.0	83.4	(27)
		MCDB	86.9	87.1	86.4	83.2	79.9	77.2	74.7	76.5	79.2	81.4	84.2	86.3	(28)
	2%	WB	81.3	81.8	80.2	78.1	74.7	72.7	70.3	71.0	73.7	77.2	78.4	81.1	(29)
		MCDB	86.0	85.7	84.9	81.8	79.0	75.8	73.8	75.0	78.2	80.9	82.2	84.9	(30)
	5%	WB	80.2	80.7	79.1	76.8	73.4	71.1	69.2	69.9	72.5	75.8	76.9	79.3	(31)
		MCDB	85.1	84.8	84.0	80.7	77.7	75.2	73.2	74.4	77.3	80.2	82.0	84.3	(32)
	10%	WB	79.2	79.7	78.1	75.5	72.2	69.7	68.3	68.5	71.5	74.5	75.6	78.1	(33)
		MCDB	84.4	84.2	83.3	80.2	77.2	74.3	72.7	73.5	76.5	79.6	81.5	83.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	9.4	10.7	12.0	14.3	14.7	15.6	15.0	11.7	9.6	8.4	8.3	(35)
		MCDBR	10.1	10.9	11.9	12.9	14.7	14.2	15.2	14.9	11.8	9.7	9.3	9.1	(36)
	5% WB	MCWBR	4.1	4.6	5.0	5.7	7.4	7.7	8.4	7.8	5.4	4.3	3.9	3.8	(37)
		MCWBR	9.2	8.9	10.2	10.8	12.0	11.6	11.6	12.0	10.1	9.1	8.6	8.6	(38)
(40) Clear-Sky Solar Irradiance	taub		0.404	0.398	0.390	0.367	0.349	0.337	0.325	0.336	0.362	0.390	0.394	0.403	(40)
		taud	2.527	2.550	2.556	2.606	2.616	2.632	2.649	2.600	2.521	2.462	2.492	2.496	(41)
	Ebn at Noon		299	297	292	287	281	279	286	292	296	296	300	299	(42)
		Edh at Noon	36	35	33	29	27	26	26	29	34	37	37	37	(43)
(44) All-Sky Solar Radiation	RadAvg		2043	1893	1756	1584	1359	1227	1375	1633	1949	2135	2191	2184	(44)
	RadStd		167	201	142	134	92	106	88	95	102	113	226	196	(45)

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	-2.03	-2.38	N/A	N/A	N/A	N/A	(46)
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page